

Ben McDowell

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EDUCATION

University of Bath

Bath, UK

B.Sc. (Hons) Computer Science and AI (Predicted First Class Honours)

Sep. 2022 – Jul 2026

- Weighted Average: 75.6% | Final Year Average: 82.3% (to date)

The College of Richard Collyer

Horsham, UK

Computer Science (A), EPQ (A*), Mathematics (A), French (A)*

Sep. 2020 – Jun 2022

PROJECTS

Semantic Grounding for Unsupervised Reinforcement Learning | *PyTorch*

Sep. 2025 – Current

- Developing a novel framework grounding unsupervised RL algorithms (DIAYN, APS, LSD) by replacing raw pixel observations with CLIP embeddings, producing semantically meaningful and transferable skills.

Hierarchical Bayesian Modelling of Component Degradation | *NumPyro, JAX*

Nov. 2025 – Dec 2025

- Built a hierarchical Bayesian model in JAX and NumPyro using HMC with NUTS sampling, applying non-centred parameterisation to resolve posterior funnel geometry and achieve robust convergence across all parameters.
- Reduced predictive RMSE from 16.94 to 6.14 over a held-out test set by incorporating material composition as a hierarchical covariate, enabling informed predictions for components with sparse operational histories.

Efficient Adaptation for Low-Resource Multilingual NMT | *Transformers, LLMs*

Mar. 2024 – May 2024

- Investigated capacity bottlenecks in massive multilingual models (The Curse of Multilinguality) and the efficacy of semi-supervised learning for low-resource languages.
- Proposed few-shot inference with pre-trained encoders (mBART) to approximate fully supervised performance without the computational cost of parallel corpora training.

EXPERIENCE

Casual Teaching Assistant

Sep. 2025 – Current

University of Bath

Bath, UK

- Guide second-year undergraduates through proofs and from-scratch implementations of core ML algorithms, covering supervised, unsupervised, and deep learning methods.

Software Engineer Intern

Jun. 2025 – Aug 2025

Jefferies

London, UK

- Designed and delivered a full-stack CI/CD tool (Angular, C# WebAPI, SQL Server) automating change ticket management, credential retrieval, and application deployment across multiple integrated APIs.
- Reduced Treasury Technology team deployment time by 60%, saving over 1.5 hours per week, with role-based access controls, real-time monitoring, and audit logging.

Business Intelligence Engineer Intern

Jun. 2024 – Jun 2025

Amazon

London, UK

- Developed forecasting models for EU carrier capacity constraints, improving average accuracy by 35% and supporting decision-making during High Volume Events.
- Built cost tracking infrastructure, including an ETL pipeline and daily-refreshing QuickSight dashboard, providing granular visibility across a >\$30M plan for all EU third-party carriers and enabling proactive management of forecasted headwinds.

Data Analyst Intern (Part-Time)

Nov. 2022 – Apr. 2024

Play Sports Network

Bath, UK

- Built an automated dashboard using the YouTube API to collect and track competitor performance data.
- Delivered commercial reports for partners covering 23+ social channels, collaborating with account managers.

TECHNICAL SKILLS

Machine Learning: PyTorch, NumPy, Scikit-learn, JAX, NumPyro, HuggingFace Transformers

Concepts: Bayesian Inference, MCMC, Variational Inference, Reinforcement Learning (SAC/PPO), LLMs

Development: Python, SQL, C#, Java, Git, Docker, AWS (SageMaker, Lambda), Linux/Bash

Mathematics: Linear Algebra (SVD, PCA), Calculus, Probability Theory, Information Theory